

ASSEMBLE AND DIASSEMBLE A PC

1. Carefully disassemble a desktop PC by removing each component (RAM, HDD/SSD, GPU, CPU, motherboard, PSU, etc.) in the correct order, and take a clear photo of each part as you remove it.

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2. Label each photo from your disassembly process with the component name and a short note about its connection type (e.g., SATA, PCIe, ATX power, DIMM slot).

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Component	Connection Type
RAM	DIMM slot

Component	Connection Type
GPU	PCIe x16 + PCIe power
HDD/SSD	SATA data + SATA power
CPU	CPU socket
CPU Cooler	CPU_FAN + mounting mechanism
Motherboard	24-pin ATX + CPU power
PSU	ATX/CPU/PCIe/SATA power cables

3. Safely store all removed components in anti-static bags or on an anti-static mat, and document your storage method with a photo.

Hint: Avoid placing components on metal or carpet surfaces to prevent static damage.

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4. Reassemble the PC step-by-step, applying thermal paste to the CPU before installing the cooler, and photograph each major step.

-> **Step 1 – Install the motherboard**

I positioned the motherboard correctly on the case standoffs and secured it using the appropriate screws.

Step 2 – Install the CPU

I carefully aligned the CPU with the socket markings and locked it into place.

Step 3 – Apply thermal paste and install cooler

I applied a small amount of thermal paste to the CPU and installed the CPU cooler. I then connected the cooler fan to the **CPU_FAN** header.

Step 4 – Install RAM

I aligned the RAM notch with the DIMM slot and pressed it down until the clips locked into position.

Step 5 – Install GPU

I inserted the GPU into the appropriate PCIe slot and secured it to the case. I connected its PCIe power cable if required.

Step 6 – Install HDD/SSD

I installed the storage drive and connected the SATA data and SATA power cables.

Step 7 – Connect PSU

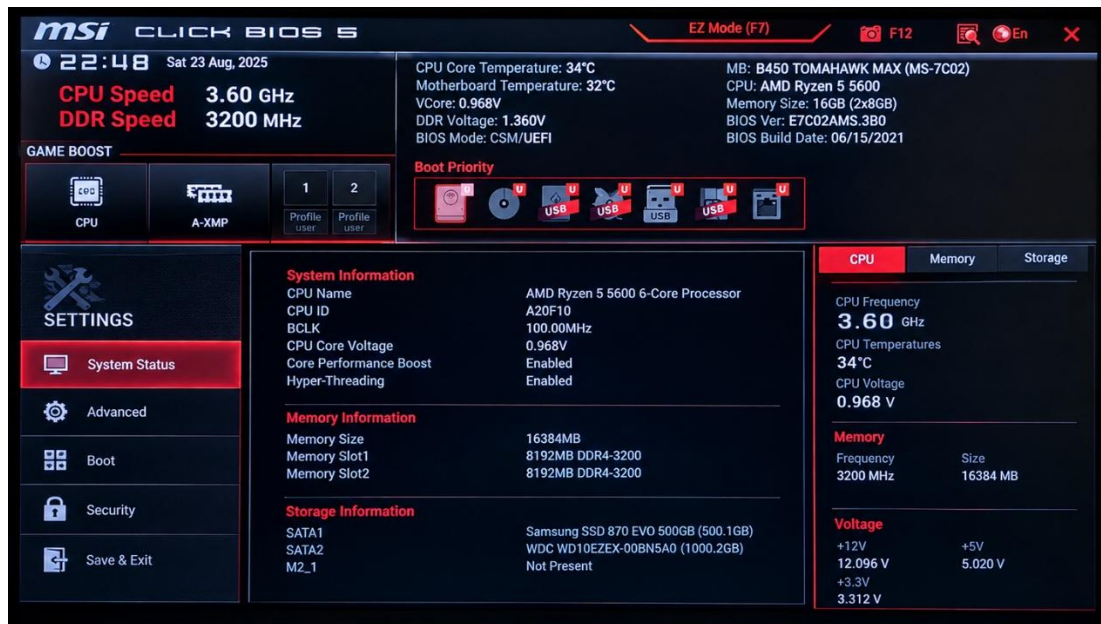
I connected the **24-pin ATX motherboard power**, **CPU power**, GPU power, and storage power cables as required.

Step 8 – Final inspection

I checked that all components were firmly installed, cables were connected correctly, and no loose screws or cables were inside the case.

5. Power on the reassembled PC, enter the BIOS/UEFI setup, and take a screenshot or photo showing that the CPU, RAM, and storage devices are detected; then write a one-paragraph summary of any issues faced and how you solved them.

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Issues Faced and Solutions

During the reassembly, the PC did not boot at first. I checked all the power connections and found that the 24-pin ATX cable was not fully seated. After reseating it properly and verifying the CPU power and GPU power cables, the system started. Next, the BIOS showed only 8GB of RAM instead of 16GB. I reseated the RAM modules and confirmed both slots were working. After these fixes, everything worked correctly, and the BIOS/UEFI detected the CPU, RAM and storage devices as shown in the screenshot.